

Samyukta Reddy Vanga

AI/ML ENGINEER

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SUMMARY

AI/ML Engineer with 5+ years of experience delivering end-to-end machine learning and deep learning solutions across healthcare and consulting domains. Proficient in Python, R, TensorFlow, PyTorch, Scikit-learn, and Big Data technologies (Spark, Kafka, Hadoop) with expertise in NLP, Generative AI, and RAG-based systems. Skilled in model deployment on AWS, Azure, and GCP using Docker and Kubernetes, ensuring scalable and reliable production applications. Experienced in building AI-driven claims optimization, fraud detection, and physician-support tools, improving accuracy and reducing manual efforts. Also developed predictive analytics, anomaly detection, and customer churn models, enabling data-driven business growth. Strong in Agile/Scrum delivery, cross-functional collaboration, and statistical Modeling, with a proven track record of driving measurable impact.

SKILLS

Methodologies: SDLC, Agile/Scrum, Waterfall
Programming Languages: Python, R, Java, C++, JavaScript
Machine Learning & AI Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, LangChain
Deep Learning: CNNs, RNNs, LSTMs, Large Language Models (LLMs)
Data Manipulation: Pandas, NumPy
Big Data Technologies: Hadoop, Spark, Apache Kafka
NLP & Generative AI: NLTK, SpaCy, BERT, GPT-3, Transformer models, AI Agents, Retrieval-Augmented Generation (RAG)
Cloud Platforms: AWS, Microsoft Azure, Google Cloud Platform (GCP)
Visualization & BI Tools: Tableau, Power BI, Streamlit
Databases: MySQL, PostgreSQL, Vector Databases
Model Deployment: Docker, Kubernetes, TensorFlow Serving
Version Control & Dev Tools: Git, GitHub, JIRA, Jenkins
Statistics & Mathematical Modelling: Statistical analysis, probability, algorithm development

EDUCATION

Master of Science in Applied Data Science **Dec 2022**
Indiana University-Purdue University, Indianapolis, IN, USA

EXPERIENCE

- Cigna, IN | AI/ML Engineer** **Sep 2022 - Current**
 - Developed and deployed AI-powered claims review models using Python, Scikit-learn, and TensorFlow, improving claims processing accuracy by 28% and reducing manual review time by 40%.
 - Implemented NLP pipelines with SpaCy, BERT, and LangChain to extract, summarize, and validate unstructured medical documents, ensuring HIPAA-compliant automation of claims adjudication.
 - Built scalable ETL pipelines with Apache Spark and Kafka for processing high-volume healthcare claims data, enabling near real-time anomaly detection and fraud prevention.
 - Led development of a Retrieval-Augmented Generation (RAG) framework with GPT-3 and vector databases for physician support, improving decision-making by providing context-aware patient history insights.
 - Deployed AI/ML models in production using Docker, Kubernetes, and TensorFlow Serving on AWS and Azure, ensuring scalability, fault tolerance, and 99.9% uptime.
 - Collaborated with cross-functional teams (data engineers, product owners, compliance) in an Agile/Scrum environment, driving sprint deliverables, backlog prioritization, and regulatory alignment.
- Infinite Infolab, India | ML Engineer** **May 2015 - March 2018**
 - Designed and trained predictive models in R and Python (Scikit-learn, TensorFlow) for customer churn prediction, improving client retention rates by 18%.
 - Developed anomaly detection pipelines using unsupervised ML algorithms (Isolation Forests, Autoencoders) to flag abnormal financial transactions in real time.
 - Implemented data preprocessing and feature engineering workflows with Pandas and NumPy, ensuring high-quality datasets for ML pipelines across multiple client projects.
 - Applied NLP models with NLTK and SpaCy to extract insights from large-scale customer feedback, driving product improvements and enhancing customer satisfaction.
 - Leveraged Big Data frameworks like Hadoop and Spark to process multi-terabyte datasets, reducing model training time from hours to minutes.
 - Built interactive BI dashboards in Tableau and Power BI to visualize model outcomes and business KPIs, enabling stakeholders to make data-driven decisions.
 - Collaborated with business analysts and domain experts in a Waterfall + Agile hybrid methodology, ensuring client-specific ML solutions were delivered on time and within scope.

CERTIFICATIONS

[Microsoft Certified Azure Fundamentals \(Az-900\)](#)